

STOCKAI v2.0 - INSTRUCTION MANUAL

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PROJECT OVERVIEW:

StockAI v2.0 is an immersive research platform for high-fidelity market simulation. It combines multi-agent modeling (LLM and Rule-based) with a real-time market kernel to simulate stock price dynamics, agent sentiment, and trading behaviors.

SYSTEM REQUIREMENTS:

- OS: Windows 10+, macOS 11+, or Linux
- Processor: Dual-core 2.0GHz+
- RAM: 4GB minimum (8GB recommended)
- Browser: Chrome, Firefox, or Edge (latest versions)
- Python 3.9.x or higher

INSTALLATION STEPS:

1. Copy the 'Code' folder to your hard drive.
2. Ensure Python is installed (check with `python --version`).
3. Follow the steps in `Software/setup_software.txt` to install dependencies and activate the virtual environment.

HOW TO RUN:

1. In your terminal, navigate to the project root.
2. Run: `python backend/run.py`
3. Open `http://127.0.0.1:8000` in your browser.

SAMPLE INPUT/OUTPUT:

- INPUT: Configure a 'Run' in the Workspace by selecting a Dataset (e.g., 'NSE_BLUECHIP') and a Scenario (e.g., 'BULL_MARKET').
- PROCESS: The simulation kernel starts processing ticks, agents begin generating trades based on their logic, and the UI updates in real-time.
- OUTPUT: Live price charts, Agent decision logs, and downloadable research snapshots (CSV/JSON).

TROUBLESHOOTING:

- "ModuleNotFoundError": Ensure you activated the virtual environment and ran `pip install -r requirements.txt`.
- "WebSocket Connection Failed": Ensure no other process is using port 8000. Refresh the page.
- "LLM Agents not responding": Check your `.env` file for valid API keys and ensure internet connectivity.
- No Data in Tables: Ensure `backend/runtime/stockai_research.db` is present in the `Code` folder.